

Actual single-cell results with the remaining limits shown

Status: numerical qualification HOLD. A converged 17-node trial has nonnegative complete film profiles and positive NMP/water transfer into the RRBO. However, its independent profile accuracy still fails the existing acceptance limit. These are **unaccepted trial values, not design results**.

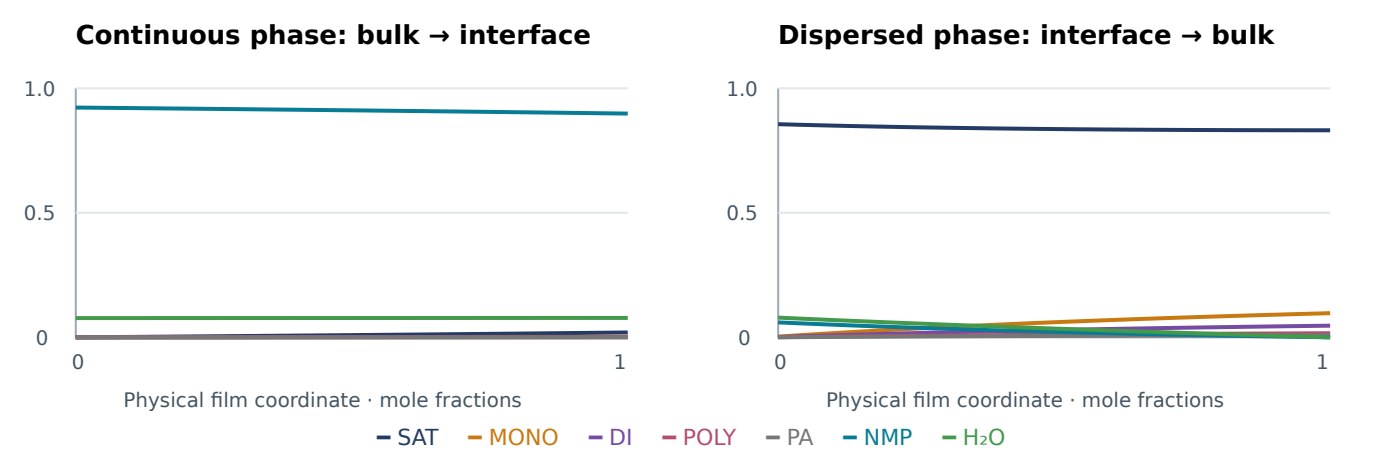
Calculated component fluxes

Positive flux is from the NMP-rich continuous phase toward the RRBO-rich dispersed phase. Units: mol/(m²·s). All seven fluxes remain free.

Component	Trial flux	Direction
SAT	-2.3983991e-03	RRBO phase → NMP phase
MONO	-3.6922303e-04	RRBO phase → NMP phase
DI	-1.7668217e-04	RRBO phase → NMP phase
POLY	-5.8353152e-05	RRBO phase → NMP phase
PA	-2.6098496e-05	RRBO phase → NMP phase
NMP	+8.4799401e-05	NMP phase → RRBO phase
H ₂ O	+2.0974838e-04	NMP phase → RRBO phase
Total	-2.7342082e-03	Sum of the seven component fluxes

Source: the completed 17-node collocation trial. Do not use these values to infer column throughput, recovery, removal, height or efficiency.

Complete calculated profiles — still unaccepted



Curves are evaluated from the authoritative Bernstein controls, not reconstructed from endpoint values. The dispersed stored water boundary is 2.50594×10^{-37} ; it appears at zero on this linear plot. The physical fresh RRBO inlet still contains exactly zero NMP and H₂O.

What passed, and what did not

Check	Observed	Required / status
Boundary-adapter derivative qualification	Previously passed	Unchanged, isolated scope only
Complete-profile nonnegativity	Both phases; all Bernstein controls nonnegative	Passed for the represented trial curves
Interface isoactivity residual	8.715e-14	$\leq 1e-7$: passed
Discrete equation residual	7.505e-13	Small; not a substitute for profile accuracy
Continuous profile constitutive defect	3.660e-08	$\leq 1e-8$: failed
Dispersed profile constitutive defect	1.300e-06	$\leq 1e-8$: failed
13→17-node common-profile change	1.145e-07	$\leq 1e-8$: failed
13→17-node scaled flux change	1.709e-10	$\leq 1e-8$: passed for this comparison only
Discrete system rank	199 / 199	Full rank; not continuum acceptance
Two successive qualifying refinements	Not established	Not passed
Candidate-matched phase stability	Not qualified	Not passed

What changed in this work

The former invalid Newton step was substantial (MONO -0.168893), not roundoff. Log-ratio coordinates prevented negative fractions but encountered extreme near-dry-boundary stiffness. Physical-coordinate collocation with feasible backtracking found the trial shown here. Exact bulk endpoints are removed from the unknowns; Bernstein evaluation avoids the old endpoint-cancellation blind spot.

The 33-node collocation refinement did not finish within its bounded execution budget. It was not accepted or extrapolated.

Phase-stability evidence: important correction

The earlier preliminary stability PASS is not valid global-search evidence: a boundary-start optimizer had not moved. The corrected, unfloored check remains blocked by two required dispersed MONO-rich optimizer failures. That check concerns older coarse interfaces, not the profiles plotted above. It does **not** establish negative TPD or physical instability.

Final high-accuracy check

The final auxiliary high-order shooting attempt did not produce a completed refined root. It encountered a bounded-execution interruption, and a post-run runner edit left an unresolved source-binding mismatch. That auxiliary evidence is excluded from qualification; it supplies no accepted profile or design result. The raw numerical record and the separate source-binding assessment are preserved.

Job A → B → C status

Existing Job-A inputs remain frozen. The local Job-B/finite-film calculation has produced trial values but is not qualified. No production Job A/B/C reruns or column continuation were performed. There is no accepted active height, physical-stage count, efficiency, final RPM or final diameter.

This remains a numerical/qualification hold—not a conclusion that the real extraction process is infeasible. The effective-resistance transport law is a modelling assumption, not measured Maxwell–Stefan transport.

Evidence references

- research-results/finite-film-cell1-lobatto-v7-final-hold.json
ba0127394224b956e13a4e96b210d4f7b68da193aae26fced8c5bf584aa4f3a7
- research-results/finite-film-cell1-lobatto-v6-focused-refinement.json
81cd9295dea730da37ddca9c9ff87451f85ba70df3f4803c37bd41c2c82bd394
- research-results/finite-film-candidate-stability-corrected-v2.json
be974df4c78bda33bb13d48b755cbc1071722c7ca9a798bddac52c2fb08f64e2
- research-results/finite-film-cell1-shooting-v8-source-binding-assessment.json
3b153e160e60ce0a955fafac4b994debe3aaddc5d5b86662c1d2996331e9e186